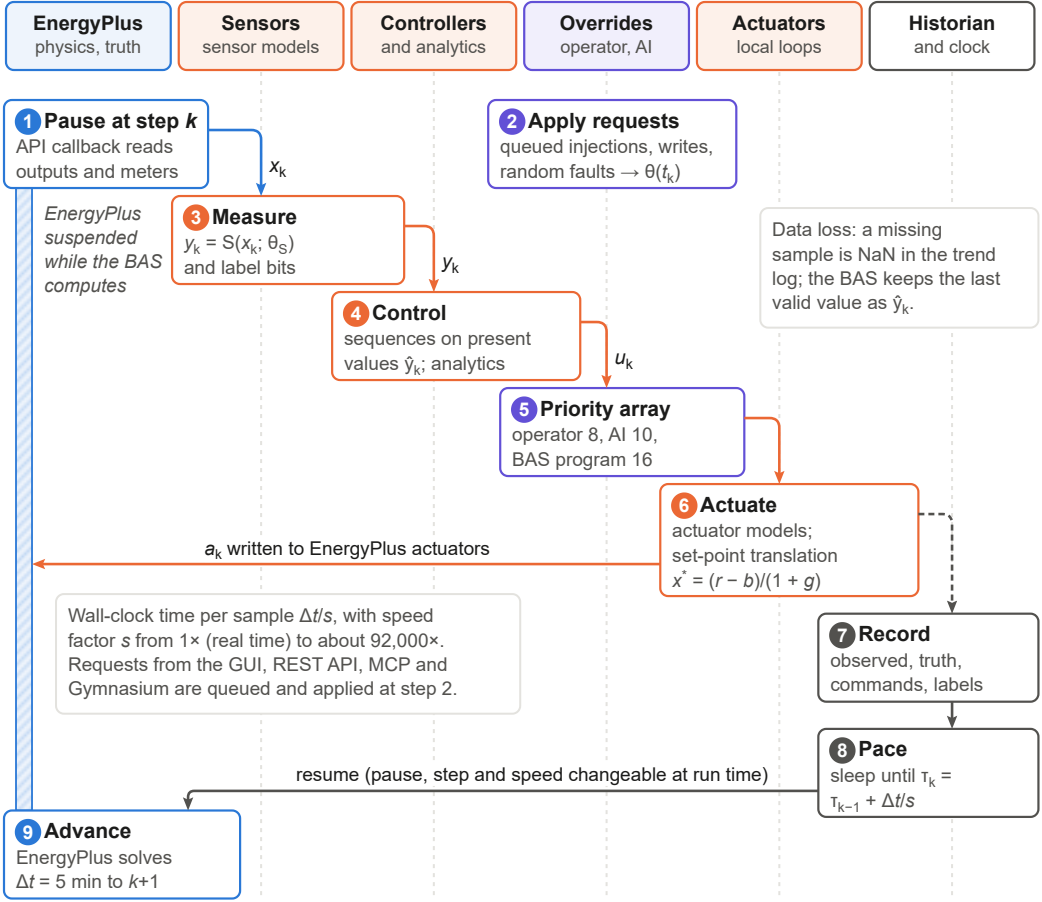

EnergyPlus
physics, truth
Sensors
sensor models
Controllers
and analytics
Overrides
operator, AI
Actuators
local loops
Historian
and clock
1 Pause at step k
API callback reads outputs and meters
x_k
2 Apply requests
queued injections, writes, random faults → θ(t_k)
EnergyPlus suspended while the BAS computes
3 Measure
y_k = S(x_k; θ_S) and label bits
y_k
Data loss: a missing sample is NaN in the trend log; the BAS keeps the last valid value as ŷ_k.
4 Control
sequences on present values ŷ_k; analytics
u_k
5 Priority array
operator 8, AI 10, BAS program 16
6 Actuate
actuator models; set-point translation
x* = (r − b)/(1 + g)
a_k written to EnergyPlus actuators
Wall-clock time per sample Δt/s, with speed factor s from 1× (real time) to about 92,000×. Requests from the GUI, REST API, MCP and Gymnasium are queued and applied at step 2.
7 Record
observed, truth, commands, labels
8 Pace
sleep until τ_k = τ_k−1 + Δt/s
resume (pause, step and speed changeable at run time)
9 Advance
EnergyPlus solves Δt = 5 min to k+1